The Ring Modulator as a Polarized Rectifier

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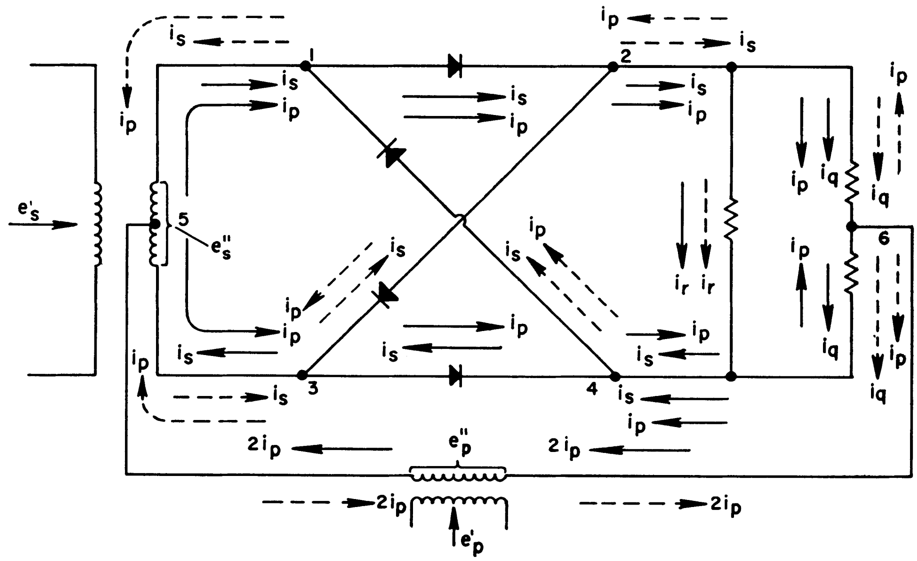
Synopsis: The performance of the ring modulator as a phase-sensitive rectifier is described qualitatively. Under the assumption of idealized conditions, and with the use of the analytical approach originally developed for the ring modulator, the polarized rectifier is described quantitatively, both for steady state and transient state. The analytical findings are checked experimentally. A differential polarized rectifier functioning as a true null detector is also discussed.

WHENEVER it becomes necessary to establish the absolute steady-state equality between two electrical phasors, a null-indicating device, or a null detector, must be used. Many such devices are in use to indicate the balance condition of bridges and, in a broader field, to permit the absolute measurement of phasor relationships.

Attention was focused early on the ring modulator, so successfully used in carrier communication, for, if instead of employing two different frequencies, i.e., the signal and carrier frequencies, the two are permitted to merge into a single measuring frequency, a new and powerful tool, the polarized rectifier, is obtained. Although the device has been previously described in the literature,[1,2] it is felt that it has been neither comprehensively treated nor given the recognition it deserves as one of great inherent usefulness. It is therefore proposed to cover this subject rather thoroughly in the following discussion.

Qualitative Aspects

The arrangement illustrated in Fig. 1(A) is diagrammatically identical to the ring modulator, although it differs functionally from the ring modulator arrangement. Fundamentally, four nonlinear resistance elements (1-2, 2-3, 3-4, 4-1), either of the vacuum-tube or the semiconductor type, are connected in series forming a "ring." (In the following, only the semiconductor type will be considered. All work was done with the Western Electric type-*D-98914* cuprous-oxide–copper varistor.) During the positive 1/2-cycle of signal e_s'' the resistance of one pair of elements is substantially reduced; during the negative 1/2-cycle, that of the other pair undergoes such a reduction, the medium producing these (exactly synchronous) reductions being an auxiliary potential, the polarizing potential e_p'', whose frequency is identically the same as that of the signal and whose amplitude is substantially in excess of that of the signal. In what manner rectification takes place will become apparent upon study of current flow for each 1/2-cycle, identified by solid and dashed arrows. Polarizing energy coupled into the circuit via an isolation transformer produces current $2i_p$ (solid), which splits at 5 into two equal components i_p. It will be assumed, for simplicity, that the resistance of an element in the blocking direction is infinite and that the characteristics of the elements are identical among each other. One component i_p thus greatly reduces the resistance of element 1-2, the other one that of 3-4: the pair of elements 1-2 and 3-4 are "unlocked" by the polarizing energy while the pair 2-3 and 4-1 becomes "forbidden" for energy transfer. Signal energy, likewise coupled by means of an isolation transformer, establishes current i_s (solid) freely flowing through 1-2, splitting into components i_q and i_r, recombining into i_s, and freely continuing through the unlocked element 4-3. For the next 1/2-cycle, polarizing current $2i_p$ (dashed) unlocks elements 2-3 and 4-1, forbidding passage through the pair 1-2 and 3-4. Signal energy, which has also undergone a synchronous phase reversal, establishes current i_s (dashed) flowing through 3-2, splitting into i_q and i_r and continuing through 4-1. It is at once realized that the direction of i_r and i_q remains invariant for either 1/2-cycle, and that hence the phenomenon of rectification takes place. A little thought will further reveal, that, if the polarity of e_p' is maintained while that of e_s' is reversed, the direction of i_r will likewise reverse: the phenomenon of passage through zero for phase reversal is guaranteed. Further study of the diagram indicates that for $e_s'=0$, no current flows through r (which represents the branch in which a d-c galvanometer



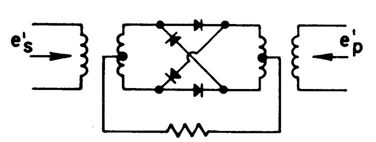


Fig. 1. Ring modulator as polarized detector

Paper 55-14, recommended by the AIEE Instruments and Measurements Committee and approved by the AIEE Committee on Technical Operations for presentation at the AIEE Winter General Meeting, New York, N. Y., January 31–February 4, 1955. Manuscript submitted March 25, 1954; made available for printing October 29, 1954.

A. J. HERMONT is with the Shell Development Company, Houston, Tex.

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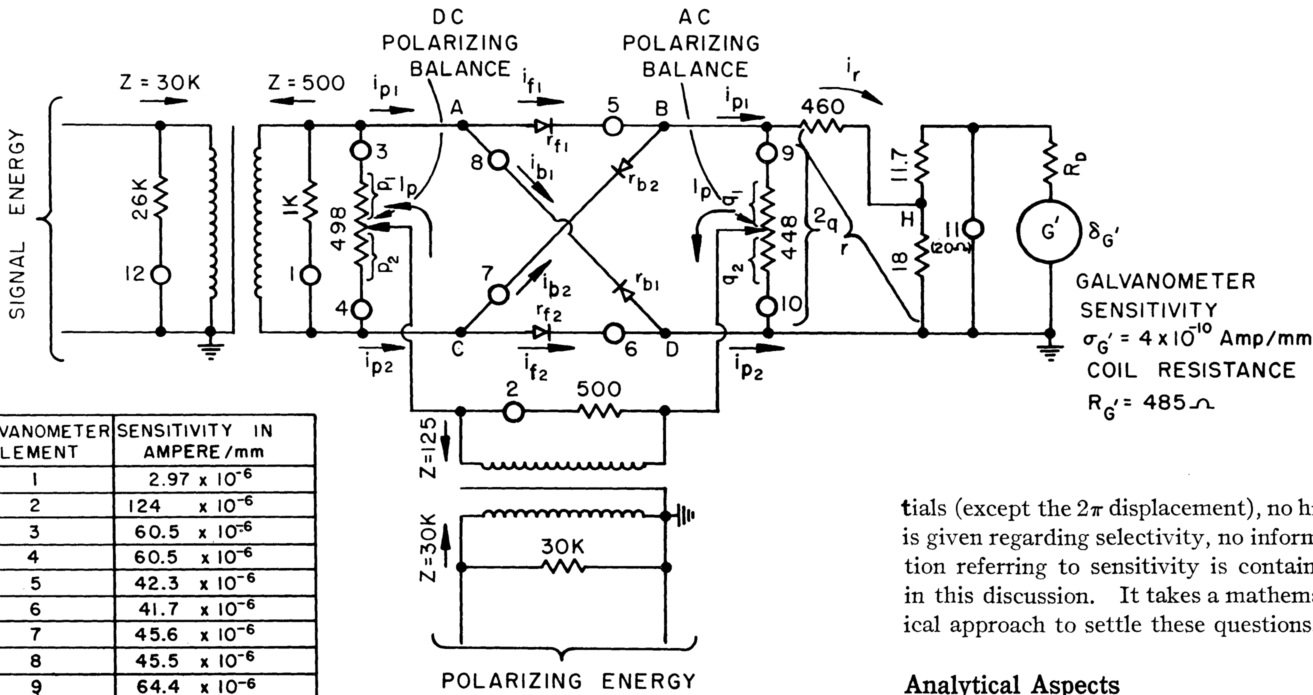


GALVANOMETER ELEMENT	SENSITIVITY IN AMPERE/mm	
1	2.97×10^{-6}	
2	124×10^{-6}	
3	60.5×10^{-6}	
4	60.5×10^{-6}	
5	42.3×10^{-6}	
6	41.7×10^{-6}	
7	45.6×10^{-6}	
8	45.5×10^{-6}	
9	64.4×10^{-6}	
10	61.9×10^{-6}	
11	1.27×10^{-6}	(LOOKING INTO POINT H)
11	$.46 \times 10^{-6}$	(ELEMENT ALONE)
12	$.85 \times 10^{-6}$	

Fig. 2. Polarized rectifier for oscillography

is placed), since polarizing energy is exactly balanced out from that branch: the phenomenon of zero output for zero input also exists. The polarized rectifier indicated in Fig. 1(B) has the same desirable properties as the one illustrated in Fig. 1(A); however, sufficient elimination of polarizing energy from the galvanometer is not possible, especially if high sensitivity is demanded.

This qualitative discussion exhibits the fundamental properties of this type of detector and puts the whole a-c null proposition on the reliable basis of a d-c null device. But no indication is provided concerning the operation of the detector as a function of phase displacement between signal and polarizing potentials (except the 2π displacement), no hint is given regarding selectivity, no information referring to sensitivity is contained in this discussion. It takes a mathematical approach to settle these questions.

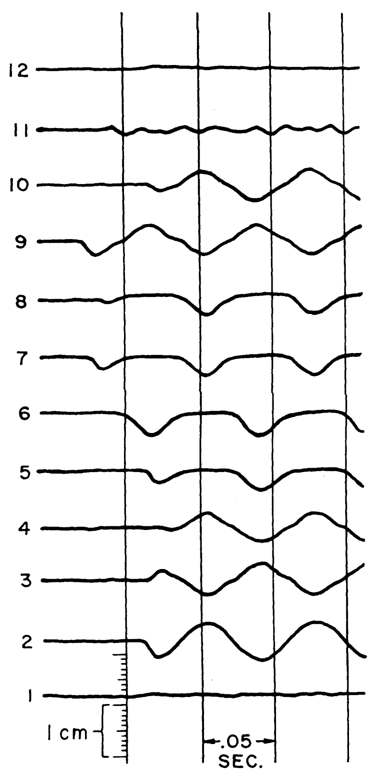
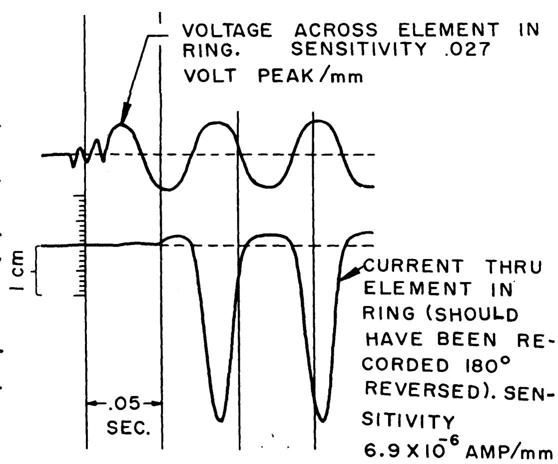


Fig. 3 (left). Exact suppression of polarizing energy

Fig. 4 (below). Voltage and current relations of a ring connected element



Analytical Aspects

The expression for the galvanometer deflection δ in terms of the harmonic signal strength E_{sm} and other physical factors has been derived in the Appendix. See equation 26.

The result may be summarized in terms of the following statements:

For idealized semiconductor specifications, the d-c output current is strictly proportional to the real component of the signal voltage with respect to the polarizing voltage. If a voltage of different frequency is superimposed on the signal, it can contribute to the d-c output current only in so far as it is an exact odd harmonic of the signal frequency: in that case the additional contribution to the output is proportional to the real component of the signal voltage and inversely proportional to the order of the harmonic; the contributions attributable to the harmonics are either aiding or bucking the contribution attributable to the fundamental.

All this reveals the paramount property that the output is not only sensitive to the real component of the signal phasor, but is also discriminative in favor of the signal frequency, so that unwanted energy is largely rejected. It is also seen very clearly that a null will obtain not only for the condition of zero input signal amplitude but also for a quadrature signal phase (with respect to the polarizing potential) and one must be on guard for this latter "fake" null.

As to the a-c output, it has already been mentioned that the fundamental frequency is completely balanced out as far as polarizing energy is concerned. In the light of the analysis this statement becomes more precise. If, for simplicity, the uncommutated signal current is assumed to be $i_{RNC} = I_{R_m} \cos m\omega t$, then the contribution to the a-c output current attributable to the nth harmonic of the commutating voltage is, by equation 7

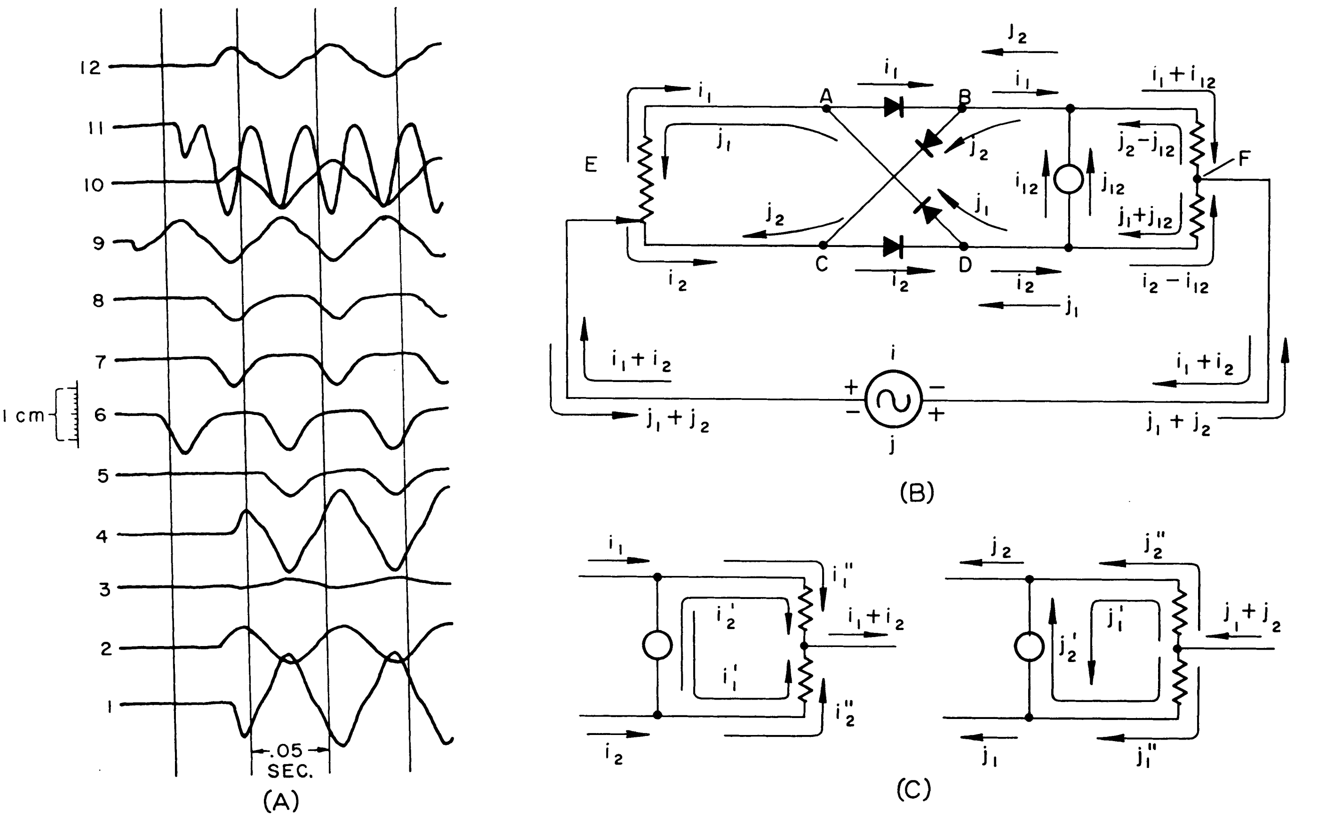


Fig. 5. D-c polarizing balance. A—Experimental results. B—Explanatory schematic. C—Formation of i_{12} and j_{12} in b

$$i_{R_{mn_c}}=\frac{4}{\pi}\frac{I_{Rm}}{n}\cos m\omega t\cos n\omega t$$
$$=\frac{2}{\pi}\frac{I_{Rm}}{n}\{\cos(m-n)\omega t+\cos(m+n)\omega t\} \quad (1)$$

If the signal frequency is exactly identical to the polarizing frequency, then $m=n=1$ and

$$i_{R_c}=\frac{2}{\pi}I_R(1+\cos 2\omega t) \quad (2)$$

so that a second harmonic (as well as a d-c component) is expected. If a second harmonic is present in the signal, i.e., $m=2n$, a fundamental component, however, will result, in addition to a third-harmonic component as

$$i_{R_{mn_c}}=\frac{2}{\pi}I_{Rm}(\cos\omega t+\cos 3\omega t) \quad (3)$$

These considerations are incorporated in the following statement: If the signal is distorted the output current contains harmonics and may contain the fundamental for the special case of a second harmonic distortion; moreover, as $m\rightarrow n$, slow beat frequencies will appear. It will be the function of the indicating device, i.e., the device capable of energy storage, to suppress these a-c components as much as possible.

The transient response is less favorable. According to equation 31 and the remark following it, every exponentially decaying step function is accompanied by a decaying train of square waves. If the polarizing frequency is low, the wave train will cause a bothersome fluctuation of the galvanometer and, unfortunately, no remedy for this defect exists.

Equation 26 represents an important result, the experimental verification of which it is proposed to study next. This will contribute to a full appreciation of the complexities involved and will shed some light on the validity of the idealizations of the functioning of the device used in the analysis.

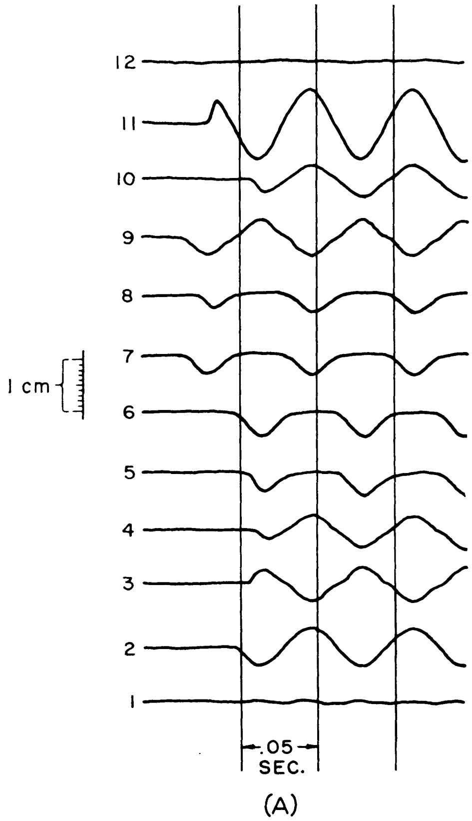
Experimental and Quantitative Aspects

Accordingly, Fig. 2, a special circuit of the general type under discussion, with suitable components for recording by means of a multichannel oscillograph has been developed. The locations of the twelve galvanometer elements (the resistance of each is about 20 ohms) are indicated by numerals 1 to 12, which correspond to the trace numbers on the oscillograms.

Fig. 3 is a tracing from an oscillogram for exact suppression of polarizing energy (frequency 14.3 cycles per second) from the output (trace 11) with zero-applied signal (trace 12). The polarizing voltage existing at the secondary of the polarizing transformer is 0.217-volt peak and is maintained throughout the tests. The d-c calibrations for the galvanometers are indicated in Fig. 2.

No fundamental is visible on trace 11 which does, however, indicate higher odd harmonics. Traces 5 and 6 show that flow of conduction currents through elements AB and CD occurs simultaneously, while conduction current flows through elements BC and DA simultaneously for the other 1/2-cycle of the polarizing voltage (traces 7 and 8 respectively). Phase reversals on traces 3 and 4, and on 9 and 10, result from the polarizing current flowing in opposite directions through resistors p_1 and p_2, and the same is true for the currents through resistors q_1 and q_2. Fundamental balance on trace 11 results for $p_1=248$ ohms and $q_1=229$ ohms, i.e., for a near-center position of the controls.

The reason for the appearance of a third harmonic on traces 3, 4, 9, 10, and 11 is fairly obvious. At all times the polarizing potential is equal to the sum of the voltage drops in p_1 and q_1, for example, and in semiconductor AB. If the polarizing potential is assumed purely sinusoidal and if the voltage across the semiconductor of the shape illustrated in Fig. 4 is subtracted from it, a voltage shape precisely as indicated by traces 3, 4, 9, and 10 results. Moreover, if the funda-



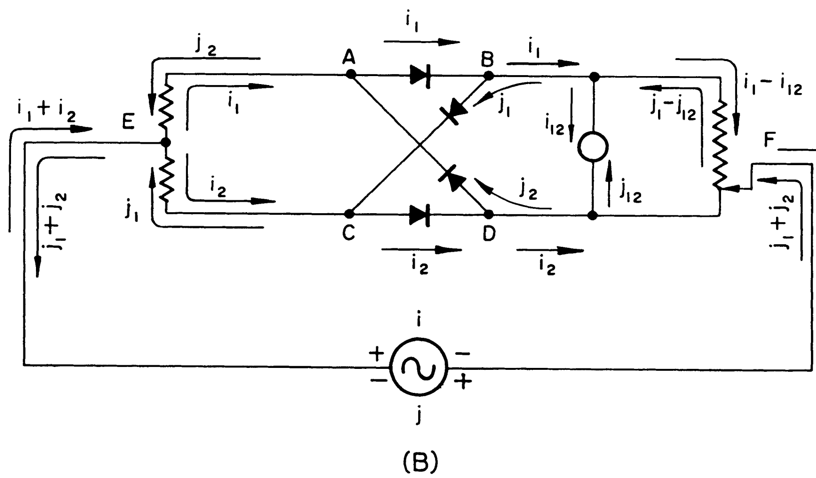


Fig. 6. A-c polarizing balance. A—Experimental results. B—Explanatory schematic

mental is balanced out, the shape indicated on trace 11 obtains.

Next, it is assumed that, still without feeding signal energy to the device, the p-balance is disturbed. Fig. 5(A) is an oscillogram for a drastic change: $p_1=498$ ohms, $p_2=0$. The result is not considered obvious, and it requires at least a qualitative analysis. A simplified network diagram is shown in Fig. 5(B), in which the polarizing energy for one 1/2-cycle feeds current i, and for the next 1/2-cycle, current j. The q-balance is shown at its center position, while the p-balance is upset as mentioned above. Thus, considering all ring elements to be identical, $i_2>i_1$, no matter how far the tap-point E is moved toward the bottom. For example, from Fig. 5(A), and with the use of the calibrations shown in Fig. 2, the maximum values of these currents are: $i_2=\delta_6\times\sigma_6=7.5\times41.7\times10^{-6}=313\times10^{-6}$ ampere and $i_1=\delta_5\times\sigma_5=5\times42.3\times10^{-6}=212\times10^{-6}$ ampere. For simplicity, it is assumed that branches AD and CB are totally blocking, so that currents i_2 and i_1 issue from points D and B respectively. Referring now, to Fig. 5(C), it is obvious that the current in the galvanometer branch becomes $i_{12}=i_2'-i_1'$ (the currents i_2 and i_1 flowing simultaneously) and, since $i_2'>i_1'$, the direction of i_{12} (Fig. 5(B)) is from D to B. For the j 1/2-cycle, the distribution of the currents is as shown, with $j_2>j_1$. Thus, the current in the galvanometer branch becomes $j_{12}=j_2'-j_1'$, in the direction of j_2', i.e., again in the direction from D to B. Thus a second-harmonic current is formed in the galvanometer branch, and this current possesses an average value. This is clearly shown on trace 11. If the p-unbalance were changed in the opposite direction, the polarity of the d-c component would reverse. These experiments indicate that the d-c component, along with the second-harmonic component, may be completely "zeroed" by an adjustment of the p-potentiometer: for then $i_2=i_1$ and $j_2=j_1$ and points B and D are at the same potential. This type of balance is loosely called the d-c polarizing balance (Fig. 2).

When next the q-balance is disturbed, no visible d-c component develops on trace 11, but fundamental frequency is evidenced, as seen from Fig. 6(A). The situation is illustrated by the schematic of Fig. 6(B), which shows the q-potentiometer drastically unbalanced (tap-off point F at bottom). Here $i_2>i_1$, and i_2 flows from D back to the generator; i_1 sends a component i_{12} through the galvanometer branch. For the j 1/2-cycle, j_2 flows directly to D while j_1 sends a component j_{12} in the opposite direction through the galvanometer branch; for identical elements $i_{12}=-j_{12}$. Cancellation, however, does not occur since these currents flow in time-quadrature; hence a near sine wave results. If the tap F is adjusted correctly, then $i_1=i_2$ and $j_1=j_2$, so that the voltage drop from F to B is equal to the one from F to D, and current in the galvanometer branch vanishes. This experiment teaches that the fundamental of the polarizing current may be completely zeroed by an adjustment of the q-potentiometer. This type of balance is referred to as the a-c polarizing balance (Fig. 2). Fig. 6(A) refers to $q_1=200$ ohms.

Before proceeding further, it is necessary to study the particular semiconductor arrangement more closely. Examination of equation 26 indicates that the d-c deflection produced by the polarized rectifier in response to an input voltage E_s depends on the quantities ρ_o and g_G, which are governed by the particulars of the circuit arrangement; it also depends on other quantities irrelevant to the present discussion, and associated with the nature of the input signal itself and with the properties of the null-indicating device. But ρ_o and g_G, by equations 16, 19, and 23 are functions of S and thus, in the last analysis, of the crucial quantities r_f and r_b by equation 17. Therefore, the question arises: what constant values should be assigned to these quantities? Of course, r_f and r_b are known to be nonlinear functions of time, but the analysis ignored this fact and assumed them to be constant. It must also be understood that the only information assumed to be available at the outset is the static current-voltage characteristic of the particular unilateral semiconductor under investigation, usually published by the manufacturer. For the Western Electric type-*D-98914* this relation is revealed for the forward 1/2-cycle and the backward 1/2-cycle by the "current" curves of Fig. 7(A), and (B) respectively. The static resistance being of no concern here, graphical differentiation of the current functions results in the dynamic-resistance functions, also shown in Fig. 7.

It should furthermore be assumed that the only information available regarding the voltage across either of the four semiconductor elements is its peak value, which is a design specification. That the voltage is not sinusoidal is illustrated by Fig. 4, which was obtained by using a d-c amplifier for the recording of the latter. It is somewhat surprising that the voltage

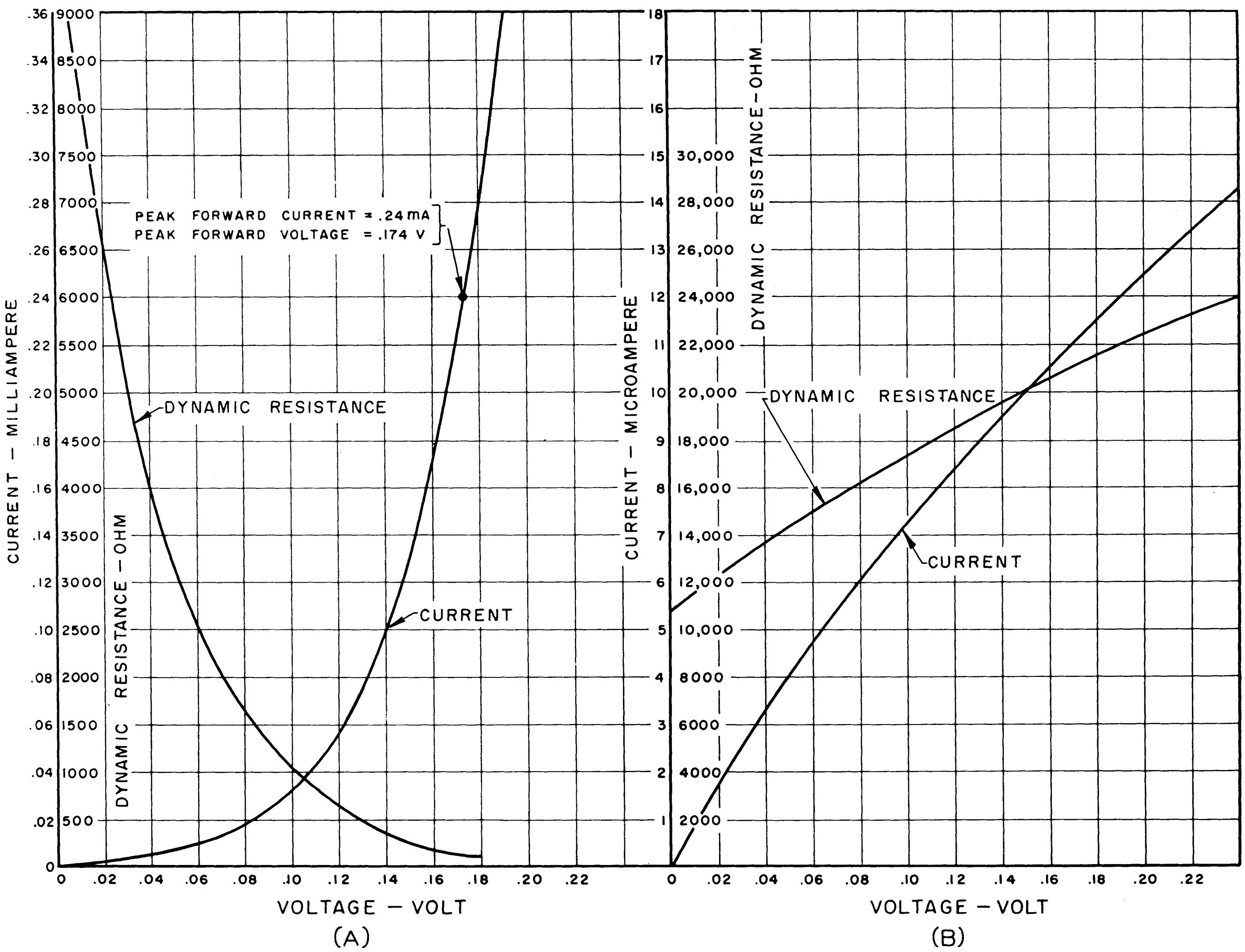


Fig. 7. Half-cycle semiconductor characteristics, Western Electric type D-98914. A—Forward. B—Backward

is so nearly sinusoidal. As an explanation of this phenomenon, it will be observed, by reference to Fig. 2, that the polarizing current I_p splits into two components i_{p1} and i_{p2}; i_{p1} splits at A into i_{b1} and i_{f1}, while i_{p2} splits at C into i_{b2} and i_{f2}, where the i_f-currents are understood to greatly exceed the i_b-currents. Assuming the potential at point D to be zero, the instantaneous potential of point A becomes $+i_{b_1}r_{b_1}$ if branch AD is considered; if branches AB and BC are considered, it becomes $+i_{f_1}r_{f_1}-i_{b_2}r_{b_2}+i_{f_2}r_{f_2}$. Thus it follows, by equating the potentials at A, that $i_{b_1}r_{b_1}+i_{b_2}r_{b_2}=i_{f_1}r_{f_1}+i_{f_2}r_{f_2}$ and, for identical elements, that

$$i_b r_b = i_f r_f \qquad (4)$$

which implies that in this arrangement there exists a "constraint" on the instantaneous voltage across an element, forcing the voltage to remain invariant for either half-cycle. This is why the voltage-time distribution is so symmetrical with respect to the time axis. With this in mind and for purposes of further simplification of the problem, it is now assumed that the voltage-time distribution across an element is strictly sinusoidal.

To restate the immediate objective: It is required to find two quantities r_f and r_b, both constant over the respective half-cycle, such that when they are introduced into the proper equations, they will yield the d-c deflection produced by the polarized rectifier as demanded by equation 26. But this is manifestly contrary to the physics of the situation, since it has been shown that the dynamic resistances are by no means constant as functions of voltage and, hence, as functions of time. The simplifying assumptions made in the analysis are nevertheless justifiable, as will be subsequently shown. These assumptions, of course, were made in order to avoid the difficulties of the nonlinear mathematics involved in the problem. It should be stressed that resort to empiricism is deliberately undertaken at this point and that all claims to rigor are relinquished as is the expectancy of perfect agreement between analysis and experiment.

Returning to the conditions of the experiment, it is seen from the calibrated Fig. 4 that the peak forward current is 0.24 milliampere (ma) and the peak forward-and-backward voltage about 0.174 volt. The distribution of the forward-and-backward dynamic resistances, plotted against a relative quarter-period time scale and based on the foregoing sinusoidal voltage-time distribution, is revealed by curves 1 of Fig. 8. The minimum forward resistance is 150 ohms, the minimum backward resistance 10,800 ohms. The question is, what effective forward-and-backward resistance may be deduced from this information?

For the forward quarter-period it is felt that the minimum resistance should play a part; also that the time-average resistance should be involved. The empirical formulation is now made that the effective forward resistance shall be the geometric mean between the time-average and the minimum resistances. Accord-

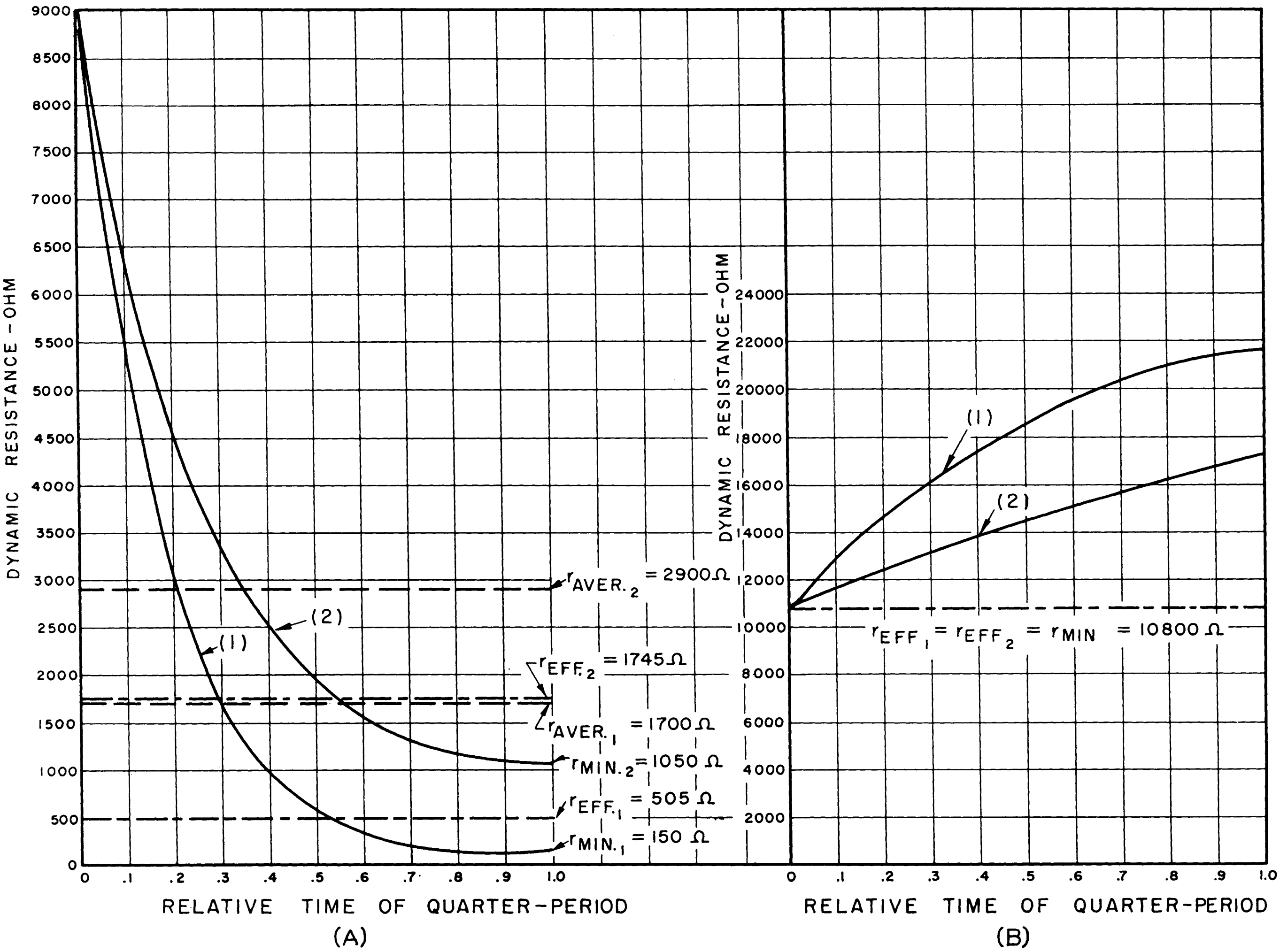


Fig. 8. Quarter-period dynamic resistance. A—Forward. B—Backward

ingly, since graphical integration of curve 1, Fig. 8(A) results in an average resistance of 1,700 ohms, the effective forward resistance becomes $r_f = 505$ ohms. For the backward quarter-period, it is felt that the rise of dynamic resistance over its value at the transition between the forward-and-backward quarter-periods should be ignored; this, too, is based on empirical procedures. Accordingly, the effective backward resistance becomes $r_b = 10{,}800$ ohms. With these values decided upon, the computation of δ from equation 26 proceeds without difficulty.

Referring to Fig. 9(A) i.e., to the case where the signal voltage is the fundamental and the phase shift is zero (or 180 degrees), equation 26 becomes $\delta = (2/\pi)(1/\sigma_G)g_G\rho_o E_s$. For the transformer indicated in Fig. 2, the experimentally determined constants are $L_1 = 387$ henrys, $R_1 = 1{,}800$ ohms, $R_2 = 52$ ohms, $\nu = 7.75$. By equation 17 $S = 2{,}335$. The fourpole resistance R_F may be computed from equation 19 if $R = 2qr//2q+r = 448\times 471.5/919.5 = 229.7$ ohms is substituted, with the result $R_F = 1{,}145$ ohms. Thus the total load resistance R_o becomes 257.5 ohms (parallel combination of R_F, 1,000 ohms, 498 ohms).

Therefore, by equation 16 $\rho_o = 9.82\times 10^{-2}$, when the frequency 13.4 cycles is given consideration. Equation 23 yields $g_G = 3.72\times10^{-4}$ mho, while E_s is taken from Fig. 9(A) from the deflection of the calibrated galvanometer 12, namely $E_s = \delta_{12}\sigma_{12}26\times10^3 = 7\times0.85\times10^{-6}\times26\times10^3 =$

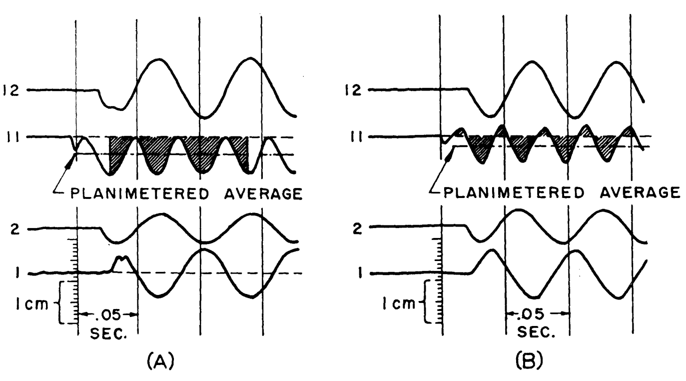
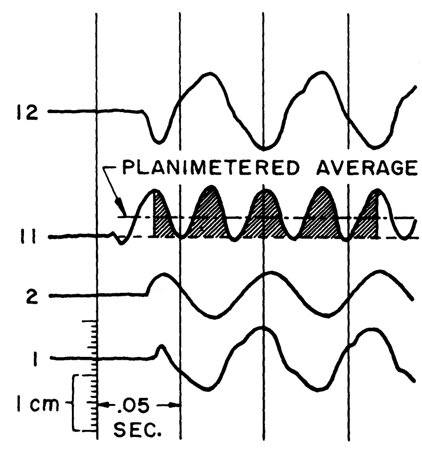


Fig. 9 (left). Signal voltage. A—In phase with polarizing voltage. B—Lagging behind polarizing voltage

Fig. 10 (right). Signal voltage containing third harmonic distortion



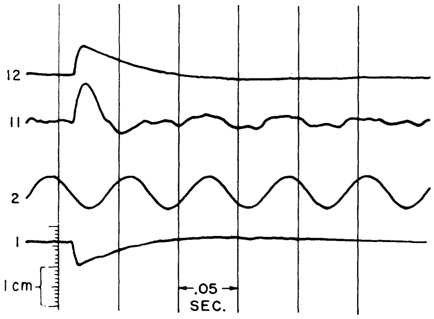


Fig. 11. Signal exponentially decaying step function

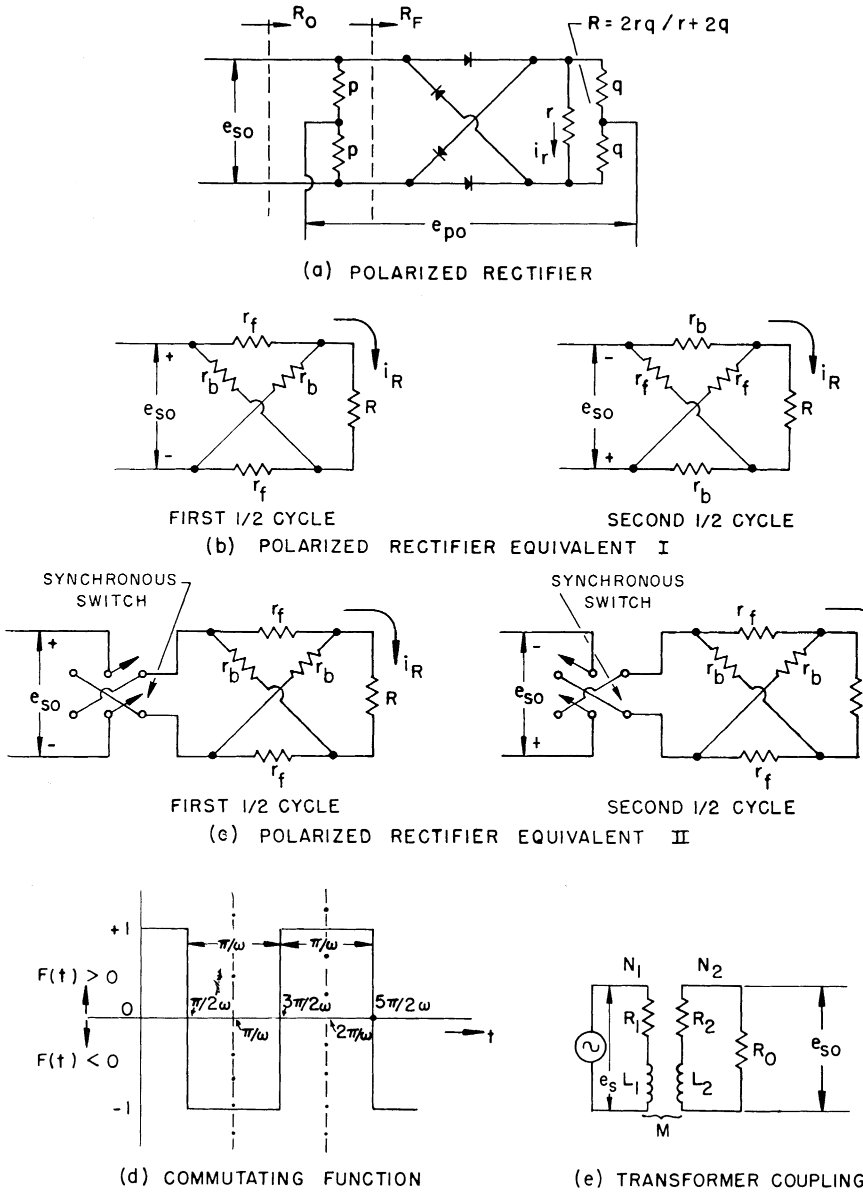


Fig. 12. Polarized rectifier analysis

0.1547-volt peak. With the galvanometer sensitivity $\sigma_G=\sigma_{11}=1.27\times10^{-6}$ finally, analysis yields $\delta_A=2/\pi\times1/1.27\times10^{-6}\times3.72\times10^{-4}\times9.82\times10^{-2}\times0.1547=$ 2.84 millimeters (mm). Next, the shaded area in Fig. 9(A) was planimetered and, as an average of seven readings, the experimental deflection by means of oscillography becomes $\delta_{EO}=4.09$ mm.

Another experimental check is provided by observing the d-c deflection of a long-period sensitive galvanometer G' and adjusting the deflection by means of a decade resistor R_D (see Fig. 2). With $R_D=5{,}000$, a galvanometer deflection $\delta_{G'}=16$ mm was observed. Taking into account the coil resistance and the sensitivity indicated in Fig. 2, the direct voltage across galvanometer 11 is computed to be 3.51×10^{-5} volt. Hence the direct current through element 11 is 1.755×10^{-6} amperes and, taking into account the sensitivity of element 11, the experimentally obtained deflection by means of a long-period galvanometer becomes $\delta_{EG'}=3.815$ mm.

Fig. 9(B) illustrates a case where phase shift is involved. Directly from the oscillogram the signal voltage (trace 1) lags behind the polarizing voltage (trace 2) by 54 degrees, if trace 1 is considered recorded 180 degrees out of phase. The oscillogram trace 12 furnishes $E_s=0.1492$-volt peak; hence $E_s\cos\varphi=0.0877$-volt peak. The other quantities remaining as in the previous case, analysis furnishes $\delta_A=1.61$ mm. The oscillographic approach yields $\delta_{EO}=2.10$ mm, while the long-period galvanometer provides $\delta_{EG'}=$ 1.91 mm.

If a strong third harmonic is present in the signal, a situation revealed by Fig. 10 results. Observe that since the signal phase (trace 1) is reversed as compared to the signal in Fig. 9(A), the d-c component on trace 11 is also reversed as expected. A Fourier analysis of the signal on trace 12 was made, the results being for $m=1$, 3,5: $E_{s1}=0.148$, $\varphi_1=-9.25$ degrees; $E_{s3}=0.0151$, $\varphi_3=+77.38$ degrees; $E_{s5}=$ 0.00638, $\varphi_5=-80.37$ degrees. If for the higher harmonics the asymptotic value of $\rho_{om}=0.0983$ is used

$$\sum_{m=1}^{m\,\mathrm{odd}}\frac{E_{sm}\rho_{om}}{m}\cos\varphi_m=0.148\times9.82\times10^{-2}\times0.9871-\frac{1}{3}\times0.0151\times9.83\times10^{-2}\times0.2181+\frac{1}{5}\times0.00638\times9.83\times10^{-2}\times0.1668=+0.014263$$

so that $\delta_A=2.66$ mm. The oscillogram yields $\delta_{EO}=3.84$ mm and the long period galvanometer $\delta_{EG'}=3.82$ mm.

If the results observed by means of the long-period galvanometer are considered correct, it must be concluded that the analytic results are at the most about 30 per cent lower than the true ones, i.e., they are on the conservative side.

Fig. 11 illustrates the response to an exponentially decaying step function, which is seen to be in complete accord with equation 31.

All these results are very gratifying, for they prove that the operation of the device is clearly understood and that the idealizing assumptions are justified in view of the close agreement between theory and experiment.

Principles of the Differential Polarized Rectifier

It is possible to interconnect two polarized rectifiers of the type shown in Figs. 1 and 12 in such a manner that if each is fed by a signal and a corresponding auxiliary signal (polarizing voltage), the detecting device (d-c galvanometer) will

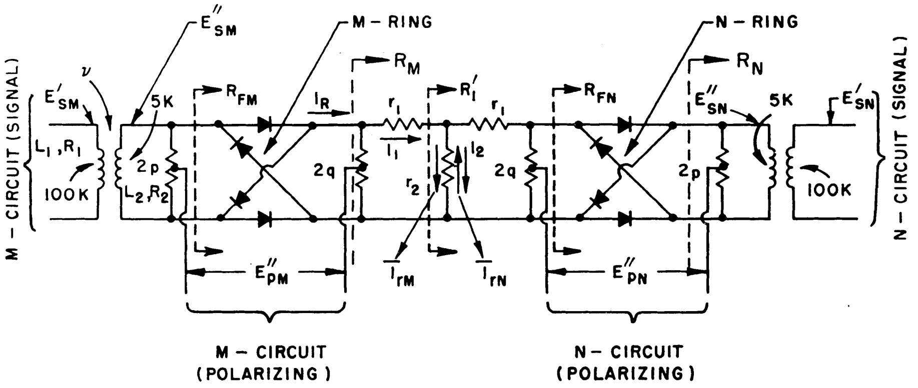


Fig. 13. Differential polarized rectifier

indicate zero current under conditions to be described.

Considering the schematic diagram shown in Fig. 13, signals E_{sM}' and E_{sN}' coming from two signal channels M and N, act upon the primaries of two signal transformers, establishing E_{sM}'' and E_{sN}'' respectively, across the secondaries of these transformers. Auxiliary voltages E_{pM}'' and E_{pN}'', supplied by polarizing channels M and N, are established at the (adjustable) mid-points of the respective resistors $2p$ and $2q$. (All voltage values refer to the magnitudes of the voltage phasors.)

Assuming for simplicity that the signals E_s contain only the fundamental frequency, equation 25, as far as the m-side is concerned, may be interpreted to mean that a direct current $I_{rM}=K_M E_{sM}'$ cos φ_M flows through resistance r_2 in the assumed direction indicated in Fig. 13. The phase angle φ_M is understood to be the one between the phasors e_{sM}'' and e_{pM}'' Assuming now that the polarity of the voltages applied to the N-side is the same as that of the corresponding voltages of the M-side and that the ring configurations are connected as indicated, it becomes obvious that $I_{rN}=K_N E_{sM}'$ cos φ_N will flow through r_2 in the opposite direction. Thus $I_r=I_{rM}-I_{rN}=0$ only if

$$K_M=K_N$$

and

$$E_{sM}' \cos \varphi_M = E_{sN}' \cos \varphi_N$$

That such a differential polarized rectifier may be employed as a basis for a true null indicator is described elsewhere.[3] It will now be shown numerically how the analysis developed in the Appendix can be utilized to determine the minimum value E_s' (say E_{sM}') required under special conditions.

Suppose the peak voltage across one of the elements is 0.1 volt. Then the peak forward current, according to Fig. 7(A) is only 0.032 milliampere and the minimum dynamic resistance is 1,050 ohms. The forward-and-backward dynamic resistances, plotted against the relative quarter-period time scales, are shown as curves 2 in Fig. 8. For the forward quarter-cycle the time-average dynamic resistance becomes 2,900 ohms and thus the effective dynamic resistance computes to 1,745 ohms. The effective dynamic resistance for the backward quarter-cycle is, as previously, 10,800 ohms. Hence, by equation 17, $S=4{,}350$ ohms. Referring to Fig. 13 it becomes clear that equation 23 may be used if r is replaced by (r_1+R_1') and if the result is multiplied by (R_1'/r_2).

The numerical values assumed in connection with Fig. 13 are those arising from the circuit described elsewhere[3] and are kept here.

Considering the N-ring terminated in $2p=5{,}000$ ohms and a 5,000-ohm transformer impedance in parallel, i.e., in $R_N=2{,}500$ ohms equation 19 yields $R_{FN}=3{,}940$ ohms. Thus, with $2q=5{,}500$ ohms, $r_1=5{,}000$ ohms and $r_2=20{,}500$ ohms, $R_1'=5{,}380$ ohms is obtained. Hence, equation 23 should be multiplied by $(5{,}380/20{,}500)=0.2625$, and $r=10{,}380$ should be used. Thereupon $g_G=0.994\times 10^{-5}$ mho. For the calculation of ρ_0 from equation 16, the M-ring is seen to be terminated in $R_M=3{,}590$ ohms, so that $R_{FM}=4{,}200$ ohms and $R_0=2{,}280$ ohms. If the frequency is taken as 10 cycles per second and the experimentally obtained transformer constants are $L_1=19{,}510$ henrys, $R_1=31{,}000$ ohms, $R_2=2{,}150$ ohms, and $\nu=4$, there results $\rho_o=0.08975$.

Next, it will be assumed that detection of $\pm 1/2$ mm is possible. Then, to detect variations with an error of 1 per cent, the deflection should be $\delta_{\min}=50$ mm. Therefore equation 26 yields the minimum required voltage E_s' at the primary of the transformer

$$E_s'=\frac{\delta_{\min}\,\sigma_G}{\left(\dfrac{2}{\pi}\right)g_G\rho_0}=0.0346 \text{ volt peak}$$

if $\sigma_G=4\times 10^{-10}$ ampere per mm is further assumed.

Conclusion

It has been shown by oscillographic technique as well as by direct observation of a long-period d-c galvanometer that the analytical expression derived for the d-c response of a polarized rectifier (involving errors sufficiently small) represents a reliable result. This was demonstrated for various types of signals. However, it is necessary to follow some empirical rules in numerical work, unless an unwarranted complexity involving nonlinear mathematics is introduced. The analytically obtained transient response is also corroborated by experiment. It is therefore felt that this discussion adds materially to a better understanding of the mechanism of operation of the polarized rectifier of the ring modulator type.

Appendix. Communication, Galvanometer, and Other Physical Factors

In the polarizing rectifier, Fig. 12(A), the polarizing circuit may be omitted entirely if only its effect on the ring is preserved. In the equivalent circuits *I*, Fig. 12(B), the semiconductor lattice is replaced by a pair of relatively low resistors r_f (forward or conducting) and a pair of relatively high resistors r_b (backward or blocking), assumed to be constant during the entire 1/2-cycle and changing location (within the lattice) abruptly as the signal voltage e_{so} changes polarity between 1/2-cycles. For either 1/2-cycle, signal current flows mainly through r_f resulting in the invariance of the direction of i_R. The polarizing voltage e_{po} in conjunction with the ring thus acts exactly as if a double-pole double-throw switch were operated synchronously, commutating the input voltage e_{so} but leaving the lattice intact: this is illustrated by the equivalent circuit *II*, Fig. 12(C).

This point of view leads to the mathematical formulation that it is only necessary to multiply the output current by a time function $F(t)$, the commutating function, having the shape shown in Fig. 12(D), to obtain the combined effect of the interaction of the polarizing voltage and the semiconductor ring circuit upon the signal voltage. $F(t)$ has the value $+1$ for one 1/2-period of duration π/ω and the value -1 for the next 1/2-period. This concept was first applied to modulator theory[4] and is here adapted to the polarized rectifier, which is merely a special case of the latter.

The Fourier series for the function $F(t)$ may be obtained without difficulty, the result being

$$F(t)=\frac{4}{\pi}\sum_{n=1}^{n\text{odd}}\pm\frac{\cos n\omega t}{n}=\frac{4}{\pi}\left(\cos\omega t-\frac{1}{3}\cos 3\omega t+\frac{1}{5}\cos 5\omega t\ldots\right) \quad (5)$$

Without loss of generality, the signal output current when no commutation takes place may be assumed to be

$$i_{RNC}=\sum_{m=1}^{m}I_{Rm}\cos(m\omega t+\varphi_m) \quad (6)$$

Now, to obtain the effect produced by commutation, it is necessary only to perform the operation

$$i_{RC}=i_{RNC}F(t) \quad (7)$$

for each component of the two series.

The formulation for the rectified component of i_{RC} is given in terms of an average for an infinitely long time, i.e.

$$I_{RDC}=\lim_{T\to\infty}\frac{1}{T}\int_0^T i_{RC}(t)dt \quad (8)$$

The contribution of the nth harmonic of $F(t)$ and the mth harmonic of i_{RNC} upon $I_{R_{DC}}$ is therefore

$$I_{Rnm_{DC}}=\lim_{T\to\infty}\frac{1}{T}\frac{4}{\pi}\frac{I_{Rm}}{n}\int_0^T\pm\cos n\omega t\cos\times(m\omega t+\varphi_m)dt \quad (9)$$

The definite integral is

$$J=\int_0^T\cos n\omega t(\cos m\omega t\cos\varphi_m-\sin m\omega t\sin\varphi_m)dt=\frac{1}{2}\cos\varphi_m\times\left(\frac{\sin(m-n)\omega t}{\omega(m-n)}+\frac{\sin(m+n)\omega t}{\omega(m+n)}\right)_0^T+\frac{1}{2}\sin\varphi_m\left(\frac{\cos(m-n)\omega t}{\omega(m-n)}+\frac{\cos(m+n)\omega t}{\omega(m+n)}\right)_0^T \quad (10)$$

Since the sin- and cos-functions have a bound at ± 1, examination of equation 10 reveals that no matter how closely $m\to n$. but as long as $m\neq n$, J remains $<\infty$ and hence, in the limit, equation 9 yields $I_{Rnm_{DC}}=0$. To obtain the limit for $m=n$ it is best to substitute first $\epsilon=m-n$ into equation 10 and then to go over to the limit $\epsilon\to 0$

$$J=\lim_{\epsilon\to 0}\left[\frac{1}{2}\cos\varphi_m\left(\frac{\sin\epsilon\omega T}{\omega\epsilon}+\frac{\sin(\epsilon+2n)\omega T}{\omega(\epsilon+2n)}\right)+\frac{1}{2}\sin\varphi_m\left(\frac{\cos\epsilon\omega T}{\omega\epsilon}+\frac{\cos(\epsilon+2n)\omega T}{\omega(\epsilon+2n)}-\frac{1}{\omega\epsilon}-\frac{1}{\omega(\epsilon+2n)}\right)\right]$$
$$=\frac{1}{2}\cos\varphi_m T+\left(\frac{1}{2}\cos\varphi_m\frac{\sin 2n\omega T}{2n\omega}+\frac{1}{2}\sin\varphi_m\frac{(\cos 2n\omega T-1)}{2n\omega}\right) \quad (11)$$

If this is substituted into equation 9, the limiting process $T\to\infty$ is completed and it is remembered that m merges with n, there obtains

$$I_{Rm_{DC}}=\pm\frac{2}{\pi}\frac{I_{Rm}}{m}\cos\varphi_m \quad (12)$$

Therefore, the total contribution to the rectified current is

$$I_{R_{DC}}=\frac{2}{\pi}\sum_{m=1}^{m\text{odd}}\frac{I_{Rm}}{m}\cos\varphi_m,\ m=1,\ -3,\ +5,\ -7,\ldots \quad (13)$$

and the direct current flowing in the galvanometer branch finally becomes

$$I_{r_{DC}}=\frac{2}{\pi}\sum_{m=1}^{m\text{odd}}\frac{I_{rm}}{m}\cos\varphi_m,\ m=1,\ -3,\ +5,\ -7,\ldots \quad (14)$$

Referring now to the transformer coupling of the signal energy, Fig. 12(E), it will be assumed that the magnetic paths which the flux must take in linking the primary, the secondary, and the mutual turns are identical: this implies zero leakage flux,[5] an idealized condition assumed to exist for the frequency range under consideration and, while never met in practice, nevertheless resulting in a fair approximation; from the standpoint of analysis the simplification is, of course, enormous. Thus, letting

$$L_1=K_1N_1^2$$
$$L_2=K_2N_2^2 \quad (15A)$$
$$M=KN_1N_2$$

where K is essentially the permeances of the flux path and N the primary and secondary turns, and stipulating that

$$K_1=K_2=K \quad (15B)$$

there results

$$L_2=L_1\frac{1}{\nu^2} \quad (15C)$$

and

$$M=\sqrt{L_1L_2}=L_1\frac{1}{\nu} \quad (15D)$$

where ν is the primary-to-secondary turns ratio. For such a transformer the magnitude of the output voltage E_{so} in terms of the input voltage E_s takes on the following expression, easily derivable from the coupled circuit of Fig. 12(E)

$$E_{so}=E_s\times\frac{\nu\omega L_1R_o}{([R_1(R_2+R_o)\nu^2]^2+(\omega L_1)^2[R_1+(R_2+R_o)\nu^2]^2)^{\frac{1}{2}}}=E_s\rho_o \quad (16)$$

Here ρ_o is the magnitude of the transmission ratio and R_o = total load resistance upon which the transformer works [see Fig. 12(A)].

The resistance looking into the terminated four-pole R_F of Fig. 12(A), is available in terms of the image characteristic resistance S and the image transfer factor, ζ from general network theory.[6] Both of these are real quantities in this particular case and are expressible in terms of the lattice elements r_f and r_b. Steady-state lattice theory is convenient and justifiable because of the simplifying assumptions of constant, nonreactive lattice elements; transient state, as far as the loaded four-pole is concerned, thus does not exist as the ring elements are commutated. For the symmetrical lattice under consideration

$$S=(r_fr_b)^{\frac{1}{2}} \quad (17)$$

$$\zeta=\ln\left(\frac{(r_b/r_f)^{\frac{1}{2}}+1}{(r_b/r_f)^{\frac{1}{2}}-1}\right);\ e^{-\zeta}=\frac{S-r_f}{S+r_f} \quad (18)$$

In terms of these, the resistance R_F becomes

$$R_F=S\frac{R+S\tanh\zeta}{S+R\tanh\zeta}=\frac{R(S^2+r_f^2)+2S^2r_f}{(S^2+r_f^2)+2Rr_f} \quad (19)$$

The load resistance R_o in equation 16 is the parallel combination of $2p$ and R_F. The situation is now equivalent to the one in which a generator with electromotive force $|e_{so}|$ and with zero internal impedance feeds the lattice. For this case the magnitude of the output current, I_R, in terms of S and ζ becomes

$$I_R=E_{so}\frac{2}{S+R}e^{-\zeta}\frac{1}{1-\frac{S-R}{S+R}e^{-2\zeta}} \quad (20)$$

If equation 18 is substituted for $e^{-\zeta}$ and $e^{-2\zeta}$, equation 20 yields

$$I_R=E_{so}\frac{S^2-r_f^2}{R(S^2+r_f^2)+2S_rf^2} \quad (21)$$

Since, furthermore

$$I_r=I_R\frac{R}{r}=I_R\frac{2q}{r+2q} \quad (22)$$

the result becomes

$$I_r=E_{so}\frac{(S^2-r_f^2)}{r(S^2+r_f^2)+S^2r_f\left(\frac{r+2q}{q}\right)}=E_{so}g_G \quad (23)$$

where g_G = transfer conductance, transferring a voltage applied to the input of the lattice to the galvanometer current.

Combining equations 16 and 23 and considering the mth harmonic

$$I_{rm}=E_{sm}\rho_{om}g_G \quad (24)$$

Substituting this into equation 14, the d-c galvanometer current is brought in the form

$$I_{r_{DC}}=\frac{2}{\pi}g_G\sum_{m=1}^{m\text{odd}}\frac{E_{sm}\rho_{om}}{m}\cos\varphi_m,\ m=1,\ -3,\ +5,\ -7,\ldots \quad (25)$$

It should be kept in mind that φ_m is understood to refer to the phase displacement existing between signal voltage e_{so} and polarizing voltage e_{po} at the secondaries of the isolation transformers, while E_{sm} refers to the signal amplitude at the primary of the signal isolation transformer.

Introducing the galvanometer sensitivity σ_G in amperes per mm, the expression for the galvanometer deflection δ in mm becomes

$$\delta=\frac{2}{\pi}\frac{1}{\sigma_G}\sum_{m=1}^{m\text{odd}}\frac{E_{sm}\rho_{om}}{m}\cos\varphi_m,\ m=1,\ -3,\ +5,\ -7,\ldots \quad (26)$$

Turning now to the transient response, it will be assumed that

$$i_{RNC}(t)=I_R\epsilon^{-\alpha t}1 \tag{27}$$

i.e., that it represents an exponentially decaying unit function. Considering only one term of the summation of equation 25, equation 7 becomes

$$i_{RC}(t)=\frac{4}{\pi}I_R\frac{1}{n}\epsilon^{-\alpha t}\cos n\omega t\ 1 \tag{28}$$

Using the appropriate identity of operational calculus transforming a time function into an equivalent operator

$$\epsilon^{-\alpha t}\cos n\omega t\ 1=\frac{p(p+\alpha)}{(p+\alpha)^2+n^2\omega^2}1 \tag{29}$$

the operational response then becomes the following

$$i_{RC}(p)=\frac{4}{\pi}I_R\frac{\dfrac{p(p+\alpha)}{n}}{(p+\alpha)^2+n^2\omega^2}1 \tag{30}$$

the solution of which, by means of the expansion theorem, is

$$i_{RC}(t)=\frac{4}{\pi}I_R\frac{1}{n}\epsilon^{-\alpha t}\cos n\omega t \tag{31}$$

which means that the response to an exponentially decaying step function is an exponentially decaying cosine wave. However, the commutating function, equation 5, is a summation of cosine waves producing a square wave. Thus the complete transient response will be an exponentially decaying square wave.

References

1. Electronic Null Detectors for Use with Impedance Bridges, **Horatio W. Lamson.** *AIEE Transactions*, vol. 66, 1947, pp. 535–40.

2. Phase Indicating Null Indicator for Bridges. *Electronics*, New York, N. Y., vol. 17, no. 8, Aug. 1944, pp. 242–60.

3. Precision Measurement of Phasors in Electrical Networks as a Function of Frequency, **A. J. Hermont.** *AIEE Transactions*, vol. 73, pt. I, Sept. 1954, pp. 365–75

4. Copper Oxide Modulators in Carrier Telephone Systems, **R. S. Carunthers.** *Bell System Technical Journal*, New York, N. Y., vol. 18, 1939, pp. 315–37.

5. Communication Networks (book), **E. A. Guillemin.** John Wiley & Sons, Inc., New York, N. Y., vol. I, 1931.

6. Communication Networks (book), **E. A. Guillemin.** John Wiley & Sons, Inc., New York, N. Y., vol. II, 1935.

A New Telegraph Serviceboard Using Electronic Circuits

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THE large increase in the use of teletypewriters in the Bell System in recent years has necessitated the development of improved arrangements for interconnecting telegraph facilities to form the networks employed by private-line services (also called leased-wire services). Greater flexibility is required in order to permit these networks to be built up and rearranged quickly to meet ever-changing customer requirements. With continuous improvements in the operating stability and reliability of telegraph line facilities, together with greatly increased demands from government and industry, private-line networks have become vastly complex.

Bell System telegraph service now uses the teletypewriter almost entirely, and manual Morse service is pratically negligible. Many networks extend over large areas of the nation, terminating in a number of cities. One employs about 85,000 miles of line facilities and another includes approximately 1,500 teletypewriter stations. These large networks are arranged for half-duplex service, which permits only one station to send at a time to all other stations on the network, and a large proportion of the stations are arranged for receiving-only operation. Such networks commonly serve government agencies, airlines, news agencies, financial institutions, and industrial organizations. Many operate at the higher teletypewriter transmitting speeds. Any false pulse or "hit" occurring on such a circuit will often interfere with transmission on the entire network unless it occurs on a 1-way branch, in which case its effect will be less serious. A disturbance of longer duration such as a steady space signal will interrupt service until located and removed. Even short interruptions to service on the more complex networks are likely to prove costly and result in serious inconvenience to the customer. Effort is necessary to reduce such occurrences to a minimum, and much development work has been directed toward improving present telegraph administrative arrangements which are performed largely at telegraph testboards. The results of this effort have culminated in the development of a new form of testboard known as a telegraph serviceboard which employs an electronic type of hub circuit to facilitate the interconnection and rearrangement of lines and loops for private-line telegraph service. (This is a multiple rather than the conventional series form of telegraph interconnection.)

Operating experience has indicated the importance of concentrating the telegraph administrative work in the hands of a minimum number of competent service attendants. This not only assures high efficiency in the performance of the work and faster restoration of service in case of trouble, but also provides economies in telegraph operating costs.

The new serviceboard accomplishes this centralization of effort by concentrating a larger work load in one position where it is easily accessible to an attendant. It employs a switchboard type of section with small strip-mounted jacks which enable an attendant to carry on his work from a seated position. The working load now handled at a single serviceboard position would occupy three or four positions of telegraph testboard. By

Paper 55-38, recommended by the AIEE Telegraph Systems Committee and approved by the AIEE Committee on Technical Operations for presentation at the AIEE Winter General Meeting, New York, N. Y., January 31-February 4, 1955. Manuscript submitted October 21, 1954; made available for printing November 17, 1954.

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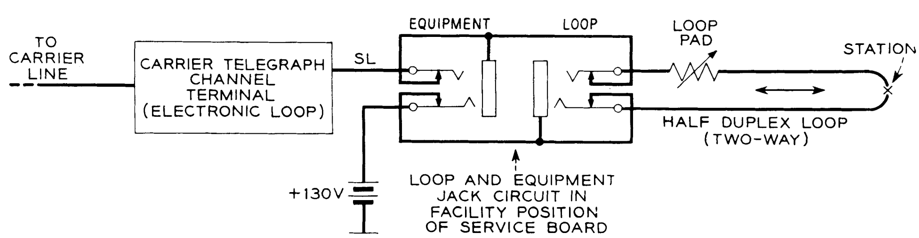

Fig. 1. Direct-leg termination of electronic carrier telegraph channel terminal in half-duplex loop